

Case C - Multi Crowd moving perpendicularly

```
clear all;
clc;

%-----Domain Parameter-----%
% Defining velocity
global INFINITE eta CFL speed_A speed_B discomfort_A discomfort_B cost_A cost_B
x_velocity_A y_velocity_A x_velocity_B y_velocity_B;

% Grid Points
global Nx; Nx = 90+6;
global Ny; Ny = 90+6;

%--Important Parameter
INFINITE = 10^12;
eta = 10^-6;
error_eta = 10^-9;
CFL = 0.5;

% To track the error for each iterations
iteration_error = [];

% Dimension of Domain
Lx = 90;
Ly = 90;

% Domain
x = linspace(0,Lx,Nx);
y = linspace(0,Ly,Ny);
[X,Y] = meshgrid(x,y);

% Space Interval
global h;
h = Lx/(Nx-6);

% Total Time
t_total = 400;

% --- Obstacle modeling ----%
global obstacle;
global P1 P2;
% Global variable for pathway width
global width; width = 10;
% Setting up the obstacle
makePathWidth(width);
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%-----Initial Condition-----%
global inflow_flux;

% Zero Density initial condition
rho_zeros = zeros(Ny,Nx);
rho = rho_zeros;    % Total Density
rho_a = rho_zeros;  % Density of A
rho_b = rho_zeros;  % Density at B

%-----Multiple Crowd Description-----%
% Rho_max and V_max
rho_max_a = 10;
rho_max_b = 10;
v_max_a = 2;
v_max_b = 2;
% Discomfort
k_a = 0.002;
k_b = 0.002;

%--Function to compute Velocity using Density
function speed = makeSpeed(rho,v_max,rho_max)
    global Nx Ny;
    speed = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            speed(j,i) = v_max*(1 - rho(j,i)/rho_max);
        end
    end
end

%--Function to compute Discomfort using Density
function discomfort = makeDiscomfort(rho,k)
    global Nx Ny;
    discomfort = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            discomfort(j,i) = k*rho(j,i)*rho(j,i);
        end
    end
end

%--Function to compute Cost function using Speed and Discomfort
function cost = makeCost(speed,discomfort)
    global Nx Ny;
    cost = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            cost(j,i) = 1/speed(j,i) + discomfort(j,i);
        end
    end
end

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end
% Function that will calculate the flux depending on the u,v and Density.
% X_velocity and y_velocity will be found using the potential.
function flux = makeFlux(rho,x_velocity,y_velocity)
    global Nx Ny;

    % Initializing Flux
    flux_x = zeros(Ny,Nx);
    flux_y = zeros(Ny,Nx);
    flux = zeros(2, Ny, Nx);

    for i = 1:1:Nx
        for j = 1:1:Ny

            flux_x(j,i) = rho(j,i) * x_velocity(j,i);
            flux_y(j,i) = rho(j,i) * y_velocity(j,i);

            flux(1,j,i) = flux_x(j,i);
            flux(2,j,i) = flux_y(j,i);
        end
    end
end

% This function will capture the flux at the exits and will return the
% updated matrix.
function matrix = captureOutflow(rho,matrix,type)
    global Nx Ny h;
    global P1 P2;
    global x_velocity_B y_velocity_A;

    total_flux = 0;

    switch type
        case 'A'
            for i = P1 : 1 : P2
                total_flux = total_flux + rho(Ny-3,i) * y_velocity_A(Ny-3,i) * h;
            end
        case 'B'
            for j = P1 : 1 : P2
                total_flux = total_flux + rho(j,Nx-3) * x_velocity_B(j,Nx-3) * h;
            end
        end

    matrix(end+1) = total_flux;
end

% Loop for the Width
for width_set = 10:10:40

    % Update the Width

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width = width_set;
% Update inflow boundary and obstacle according to width
makePathWidth(width);

% Make a new folder
width_folder = sprintf("Width_%d", width);
if ~exist(width_folder, 'dir')
    mkdir(width_folder);
end

% Loop for inflow flux
for inflow_set = 0.5:0.5:5

    % Set the inflow flux
    inflow_flux = inflow_set;

    % Zero Density initial condition
    rho_zeros = zeros(Ny,Nx);
    rho = rho_zeros;    % Total Density
    rho_a = rho_zeros;  % Density of A
    rho_b = rho_zeros;  % Density at B

    %--Pre allocating the memory
    rho_n_A = zeros(Ny,Nx);    rho_n_B = zeros(Ny,Nx);
    rho_1_A = zeros(Ny,Nx);    rho_1_B = zeros(Ny,Nx);
    rho_2_A = zeros(Ny,Nx);    rho_2_B = zeros(Ny,Nx);
    rho_next_A = zeros(Ny,Nx); rho_next_B = zeros(Ny,Nx);

    % Stopping condition
    stop_low = -1;
    stop_high = 11;

    % This will be used for plotting
    x_plot = linspace(0, Lx, Nx-6);
    y_plot = linspace(0, Ly, Ny-6);

    %-----MAIN SOLVER-----%
    t = 0;
    step = 0; % For plotting at regular interval

    % Video Capturing (Colored Density)
    file_name = fullfile(width_folder,
    sprintf("Colored_Density_L_%.2f_F_%.2f.mp4",width,inflow_flux));
    colored_rho_v = VideoWriter(file_name,"MPEG-4");
    colored_rho_v.FrameRate = 30;
    open(colored_rho_v);

    % Video Capturing (Density Contour)
    file_name = fullfile(width_folder,
    sprintf("Combined_Density_L_%.2f_F_%.2f.mp4",width,inflow_flux));

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    contour_rho_v = VideoWriter(file_name,"MPEG-4");
    contour_rho_v.FrameRate = 30;
    open(contour_rho_v);

    % Video Capturing (Potential A Contour)
    file_name = fullfile(width_folder,
sprintf("Potential_A_L_%.2f_F_%.2f.mp4",width,inflow_flux));
    contour_phiA_v = VideoWriter(file_name,"MPEG-4");
    contour_phiA_v.FrameRate = 30;
    open(contour_phiA_v);

    % Video Capturing (Potential B Contour)
    file_name = fullfile(width_folder,
sprintf("Potential_B_L_%.2f_F_%.2f.mp4",width,inflow_flux));
    contour_phiB_v = VideoWriter(file_name,"MPEG-4");
    contour_phiB_v.FrameRate = 30;
    open(contour_phiB_v);

    % Video Capturing (Shock Contour A)
    file_name = fullfile(width_folder,
sprintf("Shock_Contour_A_L_%.2f_F_%.2f.mp4",width,inflow_flux));
    contour_shockA_v = VideoWriter(file_name,"MPEG-4");
    contour_shockA_v.FrameRate = 30;
    open(contour_shockA_v);

    % Video Capturing (Shock Contour B)
    file_name = fullfile(width_folder,
sprintf("Shock_Contour_B_L_%.2f_F_%.2f.mp4",width,inflow_flux));
    contour_shockB_v = VideoWriter(file_name,"MPEG-4");
    contour_shockB_v.FrameRate = 30;
    open(contour_shockB_v);

    % This will be used to get the flux at the exits. And this will be
    % used to get the possible oscillations for low inflow flux
    outflow_flux_A = [];
    outflow_flux_B = [];
    time_array = [];

    while t < t_total

        % Coupling Term
        rho = rho_a + rho_b;

        % Computing cost for the density (rho)
        speed_A = makeSpeed(rho,v_max_a,rho_max_a);
        speed_B = makeSpeed(rho,v_max_b,rho_max_b);
        discomfort_A = makeDiscomfort(rho,k_a);
        discomfort_B = makeDiscomfort(rho,k_b);
        cost_A = makeCost(speed_A,discomfort_A);
        cost_B = makeCost(speed_B,discomfort_B);

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        %-- Calculate Initial Phi guess using Lower order Fast Sweep
        phi_first_order_A =
First_order_Godunov_Fast_Sweep_Algorithm("A",cost_A,h,INFINITE);
        phi_first_order_B =
First_order_Godunov_Fast_Sweep_Algorithm("B",cost_B,h,INFINITE);
        %-- Calculate The final phi using higher order scheme and above initial
guess
        phi_A =
weno_fastsweep("A",cost_A,h,eta,error_eta,INFINITE,phi_first_order_A,iteration_error
);
        phi_B =
weno_fastsweep("B",cost_B,h,eta,error_eta,INFINITE,phi_first_order_B,iteration_error
);

        % Extrapolation of the phi values
        phi_A = extrapolate_phi(phi_A);
        phi_B = extrapolate_phi(phi_B);

        %--Calculate the Components of Velocity
        makeComponent("A",phi_A,speed_A);
        makeComponent("B",phi_B,speed_B);

        % Updating the delta_t with proper CFL condition (Stochastic paper)
        u_max = max([abs(x_velocity_A(:)); abs(x_velocity_B(:))]);
        v_max = max([abs(y_velocity_A(:)); abs(y_velocity_B(:))]);
        %dt = CFL / ( ( u_max/h ).^(5/3) + ( v_max/h ).^(5/3) + 1e-12 );
        dt = CFL * h / ( ( u_max ) + ( v_max ) + 1e-12 );

        % ----- Making rho_n and RHS of rho_n (Crowd A and B)-----%
        rho_n_A = rho_a;
        rho_n_B = rho_b;

        semi_discritze_form_rho_n_A = makeRHS("A",rho_n_A,t);
        semi_discritze_form_rho_n_B = makeRHS("B",rho_n_B,t);

        % Check for the negative rho_n values
        if (any(rho_n_A(:) < stop_low)|| any(rho_n_B(:) < stop_low))
            break
        elseif (any(rho_n_A(:) > stop_high) || any(rho_n_B(:) > stop_high))
            break;
        end

        % Extrapolation
        rho_n_A = extrapolate_Density("A",rho_n_A);
        rho_n_B = extrapolate_Density("B",rho_n_B);

        % ----- Making the RHO_1 and RHS of rho_1 (Crowd A and B)----- %
        for i = 4:1:Nx-3
            for j = 4:1:Ny-3

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        if(~obstacle(j,i))
            rho_1_A(j,i) = rho_n_A(j,i) + dt *
semi_discritze_form_rho_n_A(j-3,i-3);
            rho_1_B(j,i) = rho_n_B(j,i) + dt *
semi_discritze_form_rho_n_B(j-3,i-3);
        else
            rho_1_A(j,i) = 0;
            rho_1_B(j,i) = 0;
        end
    end
end

% -- Extrapolation (My Intuition)
rho_1_A = extrapolate_Density("A",rho_1_A);
rho_1_B = extrapolate_Density("B",rho_1_B);

% -- RHS of discretized form using rho_1
semi_discritze_form_rho_1_A = makeRHS("A",rho_1_A,t);
semi_discritze_form_rho_1_B = makeRHS("B",rho_1_B,t);

% Check for the negative rho_1 values
if (any(rho_1_A(:) < stop_low) || any(rho_1_B(:) < stop_low))
    break
elseif (any(rho_1_A(:) > stop_high) || any(rho_1_B(:) > stop_high))
    break;
end

%----- Making the RHO_2 and RHS of RHO_2 -----%
for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        if(~obstacle(j,i))
            rho_2_A(j,i) = 3/4 * rho_n_A(j,i) + 1/4 * (rho_1_A(j,i) +
dt * semi_discritze_form_rho_1_A(j-3,i-3));
            rho_2_B(j,i) = 3/4 * rho_n_B(j,i) + 1/4 * (rho_1_B(j,i) +
dt * semi_discritze_form_rho_1_B(j-3,i-3));
        else
            rho_2_A(j,i) = 0;
            rho_2_B(j,i) = 0;
        end
    end
end

% -- Extrapolation (My Intuition)
rho_2_A = extrapolate_Density("A",rho_2_A);
rho_2_B = extrapolate_Density("B",rho_2_B);

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% -- RHS of discretized form using rho_2
semi_discritze_form_rho_2_A = makeRHS("A",rho_2_A,t);
semi_discritze_form_rho_2_B = makeRHS("B",rho_2_B,t);

% Check for the negative rho_2 values
if (any(rho_2_A(:) < stop_low)|| any(rho_2_B(:) < stop_low))
    break;
elseif (any(rho_2_A(:) > stop_high) || any(rho_2_B(:) > stop_high))
    break;
end

%----- Making the rho_next (Crowd A and B)-----%

for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        if(~obstacle(j,i))
            rho_next_A(j,i) = 1/3 * rho_n_A(j,i) + 2/3 * (rho_2_A(j,i)
+ dt * semi_discritze_form_rho_2_A(j-3,i-3));
            rho_next_B(j,i) = 1/3 * rho_n_B(j,i) + 2/3 * (rho_2_B(j,i)
+ dt * semi_discritze_form_rho_2_B(j-3,i-3));
        else
            rho_next_A(j,i) = 0;
            rho_next_B(j,i) = 0;
        end
    end
end

% -- Extrapolation (My Intuition)
rho_next_A = extrapolate_Density("A",rho_next_A);
rho_next_B = extrapolate_Density("B",rho_next_B);

%--Check if any of the density values is equal to zero
if (any(rho_next_A(:) < stop_low)|| any(rho_next_B(:) < stop_low))
    break;
elseif (any(rho_next_A(:) > stop_high) || any(rho_next_B(:) >
stop_high))
    break;
end

% Substituting back the rho value
rho_a = rho_next_A;
rho_b = rho_next_B;
rho = rho_next_A + rho_next_B;

% Now we could plot the shock waves for each density using the
% definition dell rho by dell n = gradient rho dot product
% velocity unit vector

% Calculate the density gradient for Crowd A

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[gradX_A, gradY_A] = gradient(rho_a, h, h);
shockWave_A = sqrt(gradX_A.^2 + gradY_A.^2);
% Calculate Velocity unit vector for Crowd A
velocityUnit_A = [x_velocity_A(:), y_velocity_A(:)];
velocityMagnitude_A = sqrt(sum(velocityUnit_A.^2, 2));
velocityUnit_A = velocityUnit_A ./ max(velocityMagnitude_A, 1e-12); %
Avoid division by zero
% Calculate the vector dot product
dotProduct_A = gradX_A(:) .* velocityUnit_A(:,1) + gradY_A(:) .*
velocityUnit_A(:,2);

% Calculate the density gradient for Crowd B
[gradX_B, gradY_B] = gradient(rho_b, h, h);
shockWave_B = sqrt(gradX_B.^2 + gradY_B.^2);
% Calculate Velocity unit vector for Crowd B
velocityUnit_B = [x_velocity_B(:), y_velocity_B(:)];
velocityMagnitude_B = sqrt(sum(velocityUnit_B.^2, 2));
velocityUnit_B = velocityUnit_B ./ max(velocityMagnitude_B, 1e-12); %
Avoid division by zero
% Calculate the vector dot product
dotProduct_B = gradX_B(:) .* velocityUnit_B(:,1) + gradY_B(:) .*
velocityUnit_B(:,2);

%--Plotting the rho_next;
if mod(step, 2) == 0

    figure(6);
    clf;

    % --- Custom Color Maps (White to Pure Color) --- %
    % Crowd A : White -> Bright Blue
    blueMap = [linspace(1,0,256)', linspace(1,0.4,256)',
linspace(1,1,256)'];
    % Crowd B : White -> Bright Red
    redMap = [linspace(1,1,256)', linspace(1,0,256)',
linspace(1,0,256)'];

    % Extract density matrices to avoid repetitive indexing
    density_A = rho_a(4:Ny-3, 4:Nx-3);
    density_B = rho_b(4:Ny-3, 4:Nx-3);
    % Layer 1: Crowd A (Blue)
    ax1 = axes;
    h1 = imagesc(x_plot, y_plot, density_A);
    set(gca, 'YDir', 'normal');
    set(h1, 'Interpolation', 'bilinear');

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% Apply Transparency to Crowd A so it blends with B
alphaData_A = min(density_A / 5, 1); % Fully opaque at density = 5
set(h1, 'AlphaData', alphaData_A);
axis equal tight;
xlim([0 Lx]); ylim([0 Ly]);
clim([0 10]);
colormap(ax1, blueMap);
cb1 = colorbar(ax1);
cb1.Label.String = '\rho_A';
xlabel('x'); ylabel('y');

% Layer 2: Crowd B (Red)
ax2 = axes;
h2 = imagesc(x_plot, y_plot, density_B);
set(gca, 'YDir', 'normal');
set(h2, 'Interpolation', 'bilinear');
% Apply Transparency to Crowd B
alphaData_B = min(density_B / 5, 1);
set(h2, 'AlphaData', alphaData_B);
% Make the second axis completely transparent and hide its labels
set(ax2, 'Color', 'none', 'XTick', [], 'YTick', []);
axis equal tight;
xlim([0 Lx]); ylim([0 Ly]);
clim([0 10]);
colormap(ax2, redMap);
% Match position to perfectly align the visual layers
ax2.Position = ax1.Position;
% Second colorbar setup
cb2 = colorbar(ax2);
cb2.Label.String = '\rho_B';
% Offset the second colorbar to the right so it doesn't overlap cb1
cb2.Position(1) = ax1.Position(1) + ax1.Position(3) + 0.08;
% Title & Rendering
sgtitle(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;',
width , inflow_flux, t));
drawnow;

% Save frame
frame_1 = getframe(gcf);
writeVideo(colored_rho_v, frame_1);

% Contour map

figure(7); clf;
% Density Plot
density_total = density_A + density_B;
contourf(x_plot, y_plot, density_total, 30, 'LineColor', 'none');
axis equal tight;
xlim([0 Lx]);ylim([0 Ly]);
clim([0 10]);

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colormap(gca, jet);
cb = colorbar;
cb.Label.String = '\rho_{total}';
xlabel('x');ylabel('y');
title(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;
Density Contour', width, inflow_flux, t));
drawnow;
% Save this completely separate frame
frame_2 = getframe(gcf);
writeVideo(contour_rho_v, frame_2);

figure(8); clf;
% Potential A Plot
contourf(x_plot, y_plot, phi_A(4:Ny-3, 4:Nx-3), 30, 'LineColor',
'none');

axis equal tight;
xlim([0 Lx]);ylim([0 Ly]);
colormap(gca, jet);
cb = colorbar;
cb.Label.String = '\phi_{A}';
xlabel('x');ylabel('y');
title(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;
Potential Contour - A', width, inflow_flux, t));
drawnow;
% Save this completely separate frame
frame_3 = getframe(gcf);
writeVideo(contour_phiA_v, frame_3);

figure(9); clf;
% Potential B Plot
contourf(x_plot, y_plot, phi_B(4:Ny-3, 4:Nx-3), 30, 'LineColor',
'none');

axis equal tight;
xlim([0 Lx]);ylim([0 Ly]);
colormap(gca, jet);
cb = colorbar;
cb.Label.String = '\phi_{B}';
xlabel('x');ylabel('y');
title(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;
Potential Contour - B', width, inflow_flux, t));
drawnow;
% Save this completely separate frame
frame_4 = getframe(gcf);
writeVideo(contour_phiB_v, frame_4);

figure(10); clf;
% Shock Visualization using Density Gradient Plot - A
dotProduct_A_2D = reshape(dotProduct_A, Ny, Nx);
dotProduct_A_plot = dotProduct_A_2D(4:Ny-3, 4:Nx-3);

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        contourf(x_plot, y_plot, dotProduct_A_plot, 30, 'LineColor',
'none');

        axis equal tight;
        xlim([0 Lx]);ylim([0 Ly]);
        colormap(gca, jet);
        cb = colorbar;
        cb.Label.Interpreter = 'latex'; % Enable LaTeX interpreter
        cb.Label.String = '$\frac{\partial \rho_A}{\partial n}$';
        xlabel('x');ylabel('y');
        title(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;
Shock Visualisation - A', width, inflow_flux, t));
        drawnow;
        % Save this completely separate frame
        frame_5 = getframe(gcf);
        writeVideo(contour_shockA_v, frame_5);

        figure(11); clf;
        % Shock Visualization using Density Gradient Plot - B
        dotProduct_B_2D = reshape(dotProduct_B, Ny, Nx);
        dotProduct_B_plot = dotProduct_B_2D(4:Ny-3, 4:Nx-3);
        contourf(x_plot, y_plot, dotProduct_B_plot, 30, 'LineColor',
'none');

        axis equal tight;
        xlim([0 Lx]);ylim([0 Ly]);
        colormap(gca, jet);
        cb = colorbar;
        cb.Label.Interpreter = 'latex'; % Enable LaTeX interpreter
        cb.Label.String = '$\frac{\partial \rho_B}{\partial n}$';
        xlabel('x');ylabel('y');
        title(sprintf('Path Width : %.2f; Inflow Flux : %.2f; t = %.2f;
Shock Visualisation - B', width, inflow_flux, t));
        drawnow;
        % Save this completely separate frame
        frame_6 = getframe(gcf);
        writeVideo(contour_shockB_v, frame_6);

    end

    % Capture the flux values
    outflow_flux_A = captureOutflow(rho_a,outflow_flux_A,'A');
    outflow_flux_B = captureOutflow(rho_b,outflow_flux_B,'B');
    time_array(end+1) = t;

    % Updating the time
    t = t + dt;

    step = step + 1;

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end

% Close the Video
close(colored_rho_v);
close(contour_rho_v);
close(contour_phiA_v);
close(contour_phiB_v);
close(contour_shockA_v);
close(contour_shockB_v);

% Save the CSV file with column Rho and time
data_A = [outflow_flux_A(:),time_array(:)];
data_B = [outflow_flux_B(:),time_array(:)];
file_name_A = fullfile(width_folder,
sprintf("Outflow_Flux_W_%.2f_F_%.2f_Crowd_A.csv",width,inflow_flux));
writematrix(data_A,file_name_A);
file_name_B = fullfile(width_folder,
sprintf("Outflow_Flux_W_%.2f_F_%.2f_Crowd_B.csv",width,inflow_flux));
writematrix(data_B,file_name_B);

end
end

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% Function to assign P's for width
function makePathWidth(width)
    global P1 P2;
    global obstacle;
    global Nx Ny;

    switch width
        case 10
            P1 = 43; P2 = 54;
        case 20
            P1 = 38; P2 = 59;
        case 30
            P1 = 33; P2 = 64;
        case 40
            P1 = 28; P2 = 69;
    end

    % Update the Obstacle
    obstacle = false(Ny,Nx);
    obstacle(1:P1,1:P1) = true;           % Bottom Left
    obstacle(1:P1,P2:Nx) = true;          % Bottom Right
    obstacle(P2:Ny,4:P1) = true;          % Top Left
    obstacle(P2:Ny,P2:Nx) = true;         % Top Right

end

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Components of Velocity

```
function makeComponent(type,phi,speed)
    global Nx Ny eta h x_velocity_A y_velocity_A x_velocity_B y_velocity_B;

    [phi_x, phi_y] = gradient(phi, h, h); % partial derivatives of phi in x and y
    mag_phi = sqrt(phi_x.^2 + phi_y.^2);
    mag_phi = max(mag_phi, eta);          % To be safe with denominator

    switch type
        case "A"
            for i = 1:1:Nx
                for j = 1:1:Ny
                    x_velocity_A(j,i) = speed(j,i) * (- phi_x(j,i))/mag_phi(j,i);
                    y_velocity_A(j,i) = speed(j,i) * (- phi_y(j,i))/mag_phi(j,i);
                end
            end
        case "B"
            for i = 1:1:Nx
                for j = 1:1:Ny
                    x_velocity_B(j,i) = speed(j,i) * (- phi_x(j,i))/mag_phi(j,i);
                    y_velocity_B(j,i) = speed(j,i) * (- phi_y(j,i))/mag_phi(j,i);
                end
            end
    end
end
```

First order Godunov Fast Sweep Algorithm (Older Version)

```
function phi = First_order_Godunov_Fast_Sweep_Algorithm(type,c,h,INFINITE)
    global Nx Ny obstacle P1 P2;
    phi = INFINITE*ones(Ny, Nx);    % Initial potential field

    % Exit will depend on the type of crowd
    switch type
        case "A"
            phi(Ny-3,(P1+1):(P2-1))= 0; % Top Boundary
        case "B"
            phi((P1+1):(P2-1),Nx-3) = 0; % Right Boundary
    end

    % Sweep Iterations
    for sweep = 1:20

        % Sweep 1:
        for i = 4:Nx-2
            for j = 4:Ny-2
                if ~obstacle(j, i)
                    phi = update_phi(phi, c, i, j, h);
                end
            end
        end
    end
```

```

        else
            phi(j,i) = INFINITE;
        end
    end
end

% Extrapolate the Phi
phi = extrapolatePhi(phi);

% Sweep 2:
for i = Nx-1:-1:2
    for j = 2:Ny-1
        if ~obstacle(j, i)
            phi = update_phi(phi, c, i, j, h);
        else
            phi(j,i) = INFINITE;
        end
    end
end

% Extrapolate the Phi
phi = extrapolatePhi(phi);

% Sweep 3:
for i = Nx-1:-1:2
    for j = Ny-1:-1:2
        if ~obstacle(j, i)
            phi = update_phi(phi, c, i, j, h);
        else
            phi(j,i) = INFINITE;
        end
    end
end

% Extrapolate the Phi
phi = extrapolatePhi(phi);

% Sweep 4:
for i = 2:Nx-1
    for j = Ny-1:-1:2
        if ~obstacle(j, i)
            phi = update_phi(phi, c, i, j, h);
        else
            phi(j,i) = INFINITE;
        end
    end
end

% Extrapolate the Phi
phi = extrapolatePhi(phi);

```

```

end
end

```

Subfunction: Update one grid point using Godunov scheme

```

function phi = update_phi(phi, c, i, j, dx)
    a = min(phi(j, i-1), phi(j, i+1)); % x-direction neighbors
    b = min(phi(j-1, i), phi(j+1, i)); % y-direction neighbors
    c_i = c(j, i); % Local inverse speed
    if abs(a - b) >= c_i * dx
        phi(j, i) = min(a, b) + c_i * dx;
    else
        inside = 2 * (c_i * dx)^2 - (a - b)^2;
        if inside >= 0
            phi(j, i) = (a + b + sqrt(inside)) / 2;
        end
    end
end
end

```

Extrapolate the Phi Values

```

% Function will extrapolate the phi values at the boundary
function phi = extrapolatePhi(phi)
    global Nx Ny;
    % Bottom Edge    (3 Ghost Points)
    phi(3,:) = phi(4,:);
    phi(2,:) = phi(3,:);
    phi(1,:) = phi(2,:);

    % Top Edge       (3 Ghost Points)
    phi(Ny-3,:) = phi(Ny-2,:);
    phi(Ny-1,:) = phi(Ny-2,:);
    phi(Ny,:) = phi(Ny-1,:);

    % Right Edge     (3 Ghost Points)
    phi(:,Nx-3) = phi(:,Nx-2);
    phi(:,Nx-1) = phi(:,Nx-2);
    phi(:,Nx) = phi(:,Nx-1);

    % Left Edge      (3 Ghost Points)
    phi(:,3) = phi(:,4);
    phi(:,2) = phi(:,3);
    phi(:,1) = phi(:,2);
end

```

WENO FAST SWEEP

```

function phi =
weno_fast sweep(type,c,h,eta,error_eta,INFINITE,initial_guess,iteration_error)

    global Nx Ny obstacle P1 P2;

    % Phi is initialized with infinity
    phi = initial_guess;
    phi_old = INFINITE * ones(size(phi));

    % Iterations
    iterations = 0;
    loop_safety = 0;

    % Iterations
    while (sum(abs(phi(:)-phi_old(:))) > error_eta && loop_safety<1000)

        %fprintf("Iteration = %d ",iterations);
        err = sum(abs(phi(:)-phi_old(:)));
        %fprintf("Error = %0.10f\n",err);

        phi_old = phi;
        iterations = iterations + 1;
        iteration_error(iterations) = err;
        loop_safety = loop_safety + 1;

        for sweep = 1:4
            switch sweep
                case 1
                    ix = 4:1:Nx-3; jy = 4:1:Ny-3;
                case 2
                    ix = 4:1:Nx-3; jy = Ny-3:-1:4;
                case 3
                    ix = Nx-3:-1:4; jy = 4:1:Ny-3;
                case 4
                    ix = Nx-3:-1:4; jy = Ny-3:-1:4;
            end

            for i = ix
                for j = jy
                    %if the point is outside the obstacle
                    if(~obstacle(j,i))
                        %-----Calculating the phix_min-----_%

                        r_back = (eta + (phi(j,i) - 2*phi(j,i-1) + phi(j,i-2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);
                        r_front = (eta + (phi(j,i) - 2*phi(j,i+1) + phi(j,i+2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);

                        % Calculate the w
                        w_front = 1/(1+2*(r_front^2));

```

```

        w_back = 1/(1+2*(r_back^2));

        % Dell phi by Dell x plus and minus
        phix_minus = (1-w_back)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_back*((3*phi(j,i) - 4*phi(j,i-1) + phi(j,i-2))/(2*h));
        phix_plus = (1-w_front)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_front*((-3*phi(j,i)+4*phi(j,i+1)-phi(j,i+2))/(2*h));

        phix_min = min((phi(j,i) - h*phix_minus),(phi(j,i) +
h*phix_plus));

        %-----calculating the phiy_min-----%

        r_back = (eta + (phi(j,i) - 2*phi(j-1,i) + phi(j-2,i))^2) /
(eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);
        r_front = (eta + (phi(j,i) - 2*phi(j+1,i) +
phi(j+2,i))^2) / (eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);

        % Calculate the w
        w_front = 1 / (1 + 2*(r_front^2));
        w_back = 1 / (1 + 2*(r_back^2));

        % Dell phi by Dell x plus and minus
        phiy_minus = (1-w_back) * ((phi(j+1,i)-phi(j-1,i))/(2*h)) +
w_back * ((3*phi(j,i) - 4*phi(j-1,i) + phi(j-2,i))/(2*h));
        phiy_plus = (1-w_front) * ((phi(j+1,i) - phi(j-1,i))/(2*h))
+ w_front * ((-3*phi(j,i) + 4*phi(j+1,i) - phi(j+2,i))/(2*h));

        phiy_min = min((phi(j,i) - h*phiy_minus), (phi(j,i) +
h*phiy_plus));

        if abs(phix_min - phiy_min) >= c(j,i)*h
            phi(j,i) = min(phiy_min,phix_min) + c(j,i)*h;
        else
            phi(j,i) = ((phix_min + phiy_min) + (2*(c(j,i)*h)^2 -
(phix_min - phiy_min)^2)^0.5)/2;
        end

        % Enforce the value of phi to be zero at exits depending on
the crowd

        switch type
            case "B"
                % Right Boundary
                if(i == Nx-3 && (j >= (P1+1) && j <= (P2-1)))
                    phi(j,i) = 0;
                end
            case "A"
                % Top Boundary

```

```

        if(j == Ny-3 && (i >= (P1+1) && i <= (P2-1)))
            phi(j,i) = 0;
        end
    end

    % If the point is inside the obstacle
    else
        phi(j,i) = INFINITE;
    end
end
end
end
end
end
end
end

```

Extrapolate Phi

```

function phi = extrapolate_phi(phi)
    global Ny Nx;
    % Bottom Edge (3 Ghost Points)
    phi(3,:) = phi(4,:);
    phi(2,:) = phi(3,:);
    phi(1,:) = phi(2,:);
    % Top Edge (3 Ghost Points)
    phi(Ny-2,:) = phi(Ny-3,:);
    phi(Ny-1,:) = phi(Ny-2,:);
    phi(Ny,:) = phi(Ny-1,:);
    % Right Edge (3 Ghost Points)
    phi(:,Nx-2) = phi(:,Nx-3);
    phi(:,Nx-1) = phi(:,Nx-2);
    phi(:,Nx) = phi(:,Nx-1);
    % Left Edge (3 Ghost Points)
    phi(:,3) = phi(:,4);
    phi(:,2) = phi(:,3);
    phi(:,1) = phi(:,2);
end

```

RHS of Semi Discretize Form

```

function semi_discrete_form = makeRHS(type,rho,t)
    global Nx Ny x_velocity_A y_velocity_A x_velocity_B y_velocity_B h obstacle P1
    P2 inflow_flux;
    semi_discrete_form = zeros(Ny-6,Nx-6);

    % Flux for the given rho distribution (Will depend on the type of crowd)
    switch type
        case "A"
            x_velocity = x_velocity_A;

```

```

        y_velocity = y_velocity_A;
    case "B"
        x_velocity = x_velocity_B;
        y_velocity = y_velocity_B;
    end
    flux = makeFlux(rho, x_velocity, y_velocity);
    flux_x = squeeze(flux(1,:,:));
    flux_y = squeeze(flux(2,:,:));

% Calculating the maximum speed (multiplying by 1.5 as suggested)
alpha_x = max(abs(x_velocity(:))) * 1.5;
alpha_y = max(abs(y_velocity(:))) * 1.5;

% -- Calculating the value of f+ and f- at {i+2,i+1,i,i-1,i-2,i-3} -- %
fx_negative_whole = zeros(Ny,Nx);
fx_positive_whole = zeros(Ny,Nx);
fy_negative_whole = zeros(Ny,Nx);
fy_positive_whole = zeros(Ny,Nx);

for j = 1:1:Ny
    for i=1:1:Nx
        fx_positive_whole(j,i) = 1/2 * (flux_x(j,i) + alpha_x*rho(j,i));
        fx_negative_whole(j,i) = 1/2 * (flux_x(j,i) - alpha_x*rho(j,i));
        fy_positive_whole(j,i) = 1/2 * (flux_y(j,i) + alpha_y*rho(j,i));
        fy_negative_whole(j,i) = 1/2 * (flux_y(j,i) - alpha_y*rho(j,i));
    end
end

% -- Computing the RHS of Semi Discrete form.
for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        % compute only at the points where obstacle doesn't exist
        if(~obstacle(j,i))
            %--X--%

            % fx{+}_{i+1/2,j} - Given in the paper
            fx_plus_x_half_forward = stencil( ...
                fx_positive_whole(j,i-2), ...
                fx_positive_whole(j,i-1), ...
                fx_positive_whole(j,i), ...
                fx_positive_whole(j,i+1), ...
                fx_positive_whole(j,i+2));

            % fx{+}_{i-1/2,j} - By substituting i -> (i-1) in fx{+}_{i+1/2,j}
            fx_plus_x_half_backward = stencil( ...
                fx_positive_whole(j,i-3), ...
                fx_positive_whole(j,i-2), ...
                fx_positive_whole(j,i-1), ...
                fx_positive_whole(j,i), ...

```

```

        fx_positive_whole(j,i+1));

% fx{-}_[i+1/2,j] ( mirror of fx{+}_[i+1/2,j] )
fx_minus_x_half_forward = stencil( ...
    fx_negative_whole(j,i+3), ...
    fx_negative_whole(j,i+2), ...
    fx_negative_whole(j,i+1), ...
    fx_negative_whole(j,i), ...
    fx_negative_whole(j,i-1));

% fx{-}_[i-1/2,j] ( mirror of fx{+}_[i-1/2,j] )
fx_minus_x_half_backward = stencil( ...
    fx_negative_whole(j,i+2), ...
    fx_negative_whole(j,i+1), ...
    fx_negative_whole(j,i), ...
    fx_negative_whole(j,i-1), ...
    fx_negative_whole(j,i-2));

%--Y--%

% fy{+}_[i,j+1/2]
fy_plus_y_half_forward = stencil( ...
    fy_positive_whole(j-2,i), ...
    fy_positive_whole(j-1,i), ...
    fy_positive_whole(j,i), ...
    fy_positive_whole(j+1,i), ...
    fy_positive_whole(j+2,i));

% fy{+}_[i,j-1/2]
fy_plus_y_half_backward = stencil( ...
    fy_positive_whole(j-3,i), ...
    fy_positive_whole(j-2,i), ...
    fy_positive_whole(j-1,i), ...
    fy_positive_whole(j,i), ...
    fy_positive_whole(j+1,i));

% fy{-}_[i,j+1/2]
fy_minus_y_half_forward = stencil( ...
    fy_negative_whole(j+3,i), ...
    fy_negative_whole(j+2,i), ...
    fy_negative_whole(j+1,i), ...
    fy_negative_whole(j,i), ...
    fy_negative_whole(j-1,i));

% fy{-}_[i,j-1/2]
fy_minus_y_half_backward = stencil( ...
    fy_negative_whole(j+2,i), ...
    fy_negative_whole(j+1,i), ...
    fy_negative_whole(j,i), ...
    fy_negative_whole(j-1,i), ...

```

```

        fy_negative_half(j-2,i));

        fx_positive_half = fx_plus_x_half_forward + fx_minus_x_half_forward;
        fx_negative_half = fx_plus_x_half_backward +
fx_minus_x_half_backward;
        fy_positive_half = fy_plus_y_half_forward + fy_minus_y_half_forward;
        fy_negative_half = fy_plus_y_half_backward +
fy_minus_y_half_backward;

%--Boundary Condition -- %

switch type
case "A"
    % Bottom Boundary (Inflow)
    if(j == 4 && (i >= (P1+2) && i <= (P2-2)))
        % Y flux is said to be zero at the inflow
        fx_negative_half = 0;
        % Unsteady Boundary Condition (for the x flux)
        fy_negative_half = inflow_flux;

    % Right Boundary (Wall)
    elseif(i == Nx-3 && (j >= (P1+1) && j <= (P2-1)))
        fx_positive_half = 0;
    % Left Boundary (Wall)
    elseif(i == 4 && (j >= (P1+1) && j <= (P2-1)))
        fx_negative_half = 0;
    % Top Boundary (Outflow)
    elseif(j == Ny-3 && (i >= (P1+1) && i <= (P2-1)))
        fy_positive_half = flux_y(j,i);
    end
case "B"
    % Left Boundary (Inflow)
    if(i == 4 && (j >= (P1+2) && j <= (P2-2)))
        % Y flux is said to be zero at the inflow
        fy_negative_half = 0;
        % Unsteady Boundary Condition (for the x flux)
        fx_negative_half = inflow_flux;

    % Top Boundary (Wall)
    elseif(j == Ny-3 && (i >= (P1+1) && i <= (P2-1)))
        fy_positive_half = 0;

    % Right Boundary (Outflow)
    elseif(i == Nx-3 && (j >= (P1+1) && j <= (P2-1)))
        fx_positive_half = flux_x(j,i);

    % Bottom Boundary (Wall)
    elseif(j == 4 && (i >= (P1+1) && i <= (P2-1)))
        fy_negative_half = 0;

```

```

        end

    end

    % BOUNDARY OF OBSTACLE

    % Obstacle 1 (Bottom Left Obstacle)
    if(i >= 4 && i <= P1 && j == (P1+1))
        fy_negative_half = 0; % Top Boundary
    elseif(j >= 4 && j <= P1 && i == (P1+1))
        fx_negative_half = 0; % Right Boundary
    end

    % Obstacle 2 (Top Left Obstacle)
    if(i >= 4 && i <= P1 && j == (P2-1))
        fy_positive_half = 0;
    elseif(j >= P2 && j <= Ny-3 && i == (P1+1))
        fx_negative_half = 0;
    end

    % Obstacle 3 (Bottom Right Obstacle)
    if(i >= P2 && i <= Nx-3 && j == (P1+1))
        fy_negative_half = 0;
    elseif(i == (P2-1) && j >= 4 && j <= P1)
        fx_positive_half = 0;
    end

    % Obstacle 4 (Top Right Obstacle)
    if(i >= P2 && i <= Nx-3 && j == (P2-1))
        fy_positive_half = 0;
    elseif(i == (P2-1) && j >= P2 && j <= Ny-3)
        fx_positive_half = 0;
    end

    % Computing RHS of Semi Discretized form
    semi_discrete_form(j-3,i-3) = - 1/h * (fx_positive_half -
fx_negative_half) - 1/h * (fy_positive_half - fy_negative_half);

    else
        % Else when there is an obstacle
    end

end

end

end
end

```

WENO Advection Stencil

```

% To compute f half. Order of input matters
function f = stencil(f1,f2,f3,f4,f5)

    eta = 10^-6;

```

```

% Compute S values
s_1 = ((1/3) * f1) + ((-7/6) * f2) + ((11/6) * f3);
s_2 = ((-1/6) * f2) + ((5/6) * f3) + ((1/3) * f4);
s_3 = ((1/3) * f3) + ((5/6) * f4) + ((-1/6) * f5);

% Compute Beta
beta_1 = 13/12 * (f1 - 2*f2 + f3)^2 + 1/4 * (f1 - 4*f2 + 3*f3)^2;
beta_2 = 13/12 * (f2 - 2*f3 + f4)^2 + 1/4 * (f2 - f4)^2;
beta_3 = 13/12 * (f3 - 2*f4 + f5)^2 + 1/4 * (3*f3 - 4*f4 + f5)^2;

% Defining Gamma
gamma_1 = 1/10;
gamma_2 = 3/5;
gamma_3 = 3/10;

% Computing weights bar
w_1_dash = ( gamma_1 ) ./ ((eta + beta_1)^2);
w_2_dash = ( gamma_2 ) ./ ((eta + beta_2)^2);
w_3_dash = ( gamma_3 ) ./ ((eta + beta_3)^2);

% Computing Weights
w_1 = ( w_1_dash ) / (w_1_dash + w_2_dash + w_3_dash);
w_2 = ( w_2_dash ) / (w_1_dash + w_2_dash + w_3_dash);
w_3 = ( w_3_dash ) / (w_1_dash + w_2_dash + w_3_dash);

% Computing Flux
f = w_1 * s_1 + w_2 * s_2 + w_3 * s_3;
end

```

Density Extrapolation

```

function rho = extrapolate_Density(type,rho)
    global Ny Nx;
    % Left Edge      (3 Ghost Points)
    rho(:,3) = rho(:,4);
    rho(:,2) = rho(:,3);
    rho(:,1) = rho(:,2);
    % Bottom Edge    (3 Ghost Points)
    rho(3,:) = rho(4,:);
    rho(2,:) = rho(3,:);
    rho(1,:) = rho(2,:);
    % Top Edge       (3 Ghost Points)
    rho(Ny-2,:) = rho(Ny-3,:);
    rho(Ny-1,:) = rho(Ny-2,:);
    rho(Ny,:) = rho(Ny-1,:);
    % Right Edge     (3 Ghost Points)
    rho(:,Nx-2) = rho(:,Nx-3);
    rho(:,Nx-1) = rho(:,Nx-2);

```

```
    rho(:,Nx) = rho(:,Nx-1);  
end
```